

SHERLOCK

The Riddle of the Crown Jewels — A Concise 100-Point Walkthrough, with an Opcode Census

Infocom, 1987 (`stories/sherlock.z5`, Z-machine version 5), played on the zAI-Machine interpreter with random seed 42. Final result: **100 of a possible 100**, rank *Consulting Detective*. The regalia reach Buckingham Palace at 8:55 a.m. on Monday, five minutes before the deadline. Every command below is replayed by `SherlockConsultingDetectiveTests.SherlockConsultingDetective`, which also tallies each opcode executed (see the Opcode Census at the end).

Solved during play without any outside walkthrough or hints. No game manual was supplied for this title, and the in-game HINT system was never used. The interpreter's disassembler, a parser for the V5 object table, and global/memory inspection were used to understand puzzles. This is how the following were found:

- the tide windows for the moss;
- the gems the bank guard wants;
- the dial's sound cues;
- the Monday 2:00 a.m. appointment with Akbar;
- the rope-before-blackout sequence in the lair.

The repository already holds an earlier Sherlock solution (`SherlockWalkthroughTests`). This walkthrough was solved independently of it.

Know Before You Go

- **The clock is everything.** The game runs from 5:00 a.m. Saturday to 9:00 a.m. Monday. A street move costs five minutes, and most actions cost one. Many events are tied to the clock:

- The Abbey opens at 7:00.
- Big Ben's clapper is reachable only while it strikes the hour.
- The moss under London Bridge is reachable only at low tide (Saturday 8:32–8:48 a.m. or 8:38–8:54 p.m.).
- Akbar keeps his appointment only from Monday 2:00 a.m.

`WAIT UNTIL 8:40` is the quickest way to pass time.

- **Keypresses.** Big Ben's chimes stop play with "[Press any key]", and anything that interrupts a `WAIT` asks "continue waiting? (Y/N)". The single letters K, N and Y in the command list answer these.

- **Holmes follows you.** Moriarty expects Holmes to lead, so Watson does the work. Holmes comments, and sometimes he rewards you. Reading a gem's scratched inscription with the magnifying glass while he is present earns his signet ring, the token Mycroft asks for.

- **Points come from doing things right.** You score for collecting tools, solving riddles, reviving the flower girl, haggling instead of overpaying, and summoning the right cab. Nothing in this route is wasted. It also avoids the traps that end the game early: killing the dog, breaking the ampoule near a flame, and arriving after 9:00.

- **Gems are bait.** Moriarty planted the Agra gems (ruby, sapphire, emerald, opal, topaz, garnet) as a trail. Four of them are spent bribing the bank guard, and the garnet is your ticket to the lair.

The Walkthrough

1. Baker Street (+6)

Command	What happens / why	Pts
(any key), VERBOSE, KNOCK ON DOOR, W, U, N	Mrs Hudson lets you in. Holmes broods over a syringe.	
TAKE NEWSPAPER, GIVE NEWSPAPER TO HOLMES	The notice about the closed Tower snaps him out of it. He sends for food, and the Prime Minister arrives.	+6
WAIT, N, READ PAPER, WAIT, N	The clue verse names three Abbey tombs. Holmes changes and waits outside.	

2. Equipment (+7)

Command	What happens / why	Pts
W, TAKE LAMP, TAKE GLASS, E, TAKE KNIFE, TAKE PIPE, TAKE TOBACCO, S, D, N, TAKE MATCHBOOK, S, OPEN DOOR, E, LIGHT LAMP	Each useful item scores. The lamp is needed in the fog and in dark cellars.	+7

3. Westminster Abbey (+2)

Command	What happens / why	Pts
S, S, S, SE, SE, E, E, (any key), READ SIGN, KNOCK ON DOOR, OPEN DOOR, WAIT ×6, OPEN DOOR, E	Visiting hours begin at 7:00. A verger unlocks the door.	
LOOK, S, SE, TAKE CRAYON, TAKE PACQUET, TURN OFF LAMP, NW, N	The North Cloister is the brass-rubbing centre.	+2

4. Invisible Ink (+5)

Each verse tomb holds a riddle in invisible ink on the back of its rubbing. The Evangelist Chapel's bank of candles brings it out.

Command	What happens / why	Pts
OPEN PACQUET, TAKE WHITE PAPER, PUT WHITE PAPER ON TOMB, RUB WHITE PAPER WITH CRAYON, TAKE WHITE PAPER	Newton's tomb in the Nave ("the apple ... gravity").	
NE, E, HOLD WHITE PAPER OVER CANDLES, TURN WHITE PAPER OVER	London Bridge riddle. "Subtract the conquest from the fire" gives 1666 – 1066 = 600, the safety deposit box.	+5
S, E, (yellow paper on Elizabeth's tomb), S, W, (blue paper on HENRY'S TOMB), E, NW, N, (heat and turn both)	Hickory Dickory Dock points to Big Ben. Twinkle, twinkle, son of Mars points to Nelson. Holmes names Moriarty.	

5. Nelson's Ruby (+7)

Command	What happens / why	Pts
W, SW, W, NE, W, HAGGLE WITH VENDOR ×2, BUY TELESCOPE	Haggling brings the price down to 19 shillings, leaving you a shilling in change.	+1
E, N, N, LOOK AT STATUE THROUGH TELESCOPE	A ruby sits in Nelson's blind eye.	
N, E, E, S, W, ASK SHERMAN FOR PIGEON	The half-trained pigeon only likes red things.	+1
E, N, W, W, S, LOOK AT RUBY THROUGH TELESCOPE, SHOW RUBY TO PIGEON, PIGEON, GET RUBY, RELEASE PIGEON		
N, E, E, S, W, ASK SHERMAN ABOUT RUBY	The pigeon flew home with it.	+5

6. Wiggins (+1)

Command	What happens / why	Pts
E, N, E, GIVE SHILLING TO WIGGINS, WIGGINS, STEAL KEYS	"Cor blimey, that bird isn't 'alf built." You get the master key to the safety deposit boxes.	+1

7. Big Ben (+5)

Command	What happens / why	Pts
W, W, W, S, S, S, SE, U	The Clock Tower. A sapphire is glued to the clapper over the shaft.	
(papers into packet), OPEN BAG, OPEN BLUE BOTTLE, TAKE COTTON BALLS, PUT COTTON BALLS IN EARS, WAIT ×5	At 11:00 the bell swings toward you, and the cotton protects your ears.	
TAKE SAPPHIRE, WAIT, N, TAKE SAPPHIRE	Tug on two swings, then wait out the chimes and pack the cotton away.	+5

8. Madame Tussaud's (+10)

The guard's dog smells matches and flames. Lit tobacco is exempt.

Command	What happens / why	Pts
D, NW, N, N, NE, E, S, (LOOK), N, W, SW, N, N, W	Walk to Marylebone Road.	
PUT TOBACCO IN PIPE, OPEN MATCHBOOK, TAKE MATCH, LIGHT MATCH, LIGHT TOBACCO WITH MATCH, DROP MATCHBOOK, DROP LAMP, N	Holmes spots Moriarty's cigar ash. The lit pipe lights the dark Chamber of Horrors.	+1
W, TAKE TORCH, LIGHT CLUE PAPER WITH PIPE, LIGHT TORCH WITH CLUE PAPER	The pipe can't light the torch itself, but burning paper can. "Here comes a candle."	+4
TAKE HEAD, MELT HEAD WITH TORCH, TAKE EMERALD	"Here comes a chopper": the wax head of Charles I hides the emerald.	+5

9. The Black Museum (+1)

Command	What happens / why	Pts
E, S, TAKE LAMP, TAKE MATCHBOOK, E, S, S, S, E, LIGHT LAMP, D, TAKE OAR, U, TURN OFF LAMP	Captain Bligh's oar fills the rowboat's empty oarlock.	+1

10. Covent Garden (+6)

Command	What happens / why	Pts
W, N, NE, N, DROP CRAYON, DROP MATCHBOOK, DROP TELESCOPE, TAKE OFF HAT, TAKE STETHOSCOPE, WEAR HAT, WEAR STETHOSCOPE, LISTEN TO GIRL	The fainted flower girl's heartbeat is incredibly slow.	+1
OPEN BAG, OPEN BROWN BOTTLE, TAKE YELLOW PILL, PUT YELLOW PILL IN GIRL'S MOUTH	Belladonna treats bradycardia. Eliza pins a carnation in your lapel.	+5

11. The Opal (+5)

Command	What happens / why	Pts
REMOVE STETHOSCOPE, S, E, S, GET IN BOAT, PULL CHAIN, LAUNCH BOAT, LOOK, LOOK, DROP ANCHOR	Drift under London Bridge and anchor there.	
WAIT UNTIL 8:40, Y, LIGHT LAMP, TAKE MOSS	At low tide the oddly coloured moss is within reach and crumbles to an opal.	+5
PUT BLIGH'S OAR IN OARLOCK, PULL CHAIN, LOOK, ROW N, LAUNCH BOAT, ROW W x3, ROW N, GET OUT OF BOAT	A look at the Traitors' Gate, then row back up to the Embankment.	

12. The Bank of England (+12)

Command	What happens / why	Pts
E, E, N, NW, NW, EXAMINE RUBY WITH GLASS	"Wear a carnation." Holmes hands over his heirloom ring.	+1
GIVE RUBY / SAPPHIRE / EMERALD / OPAL TO GUARD, N	The guard wants four gems before he looks the other way.	+3
TAKE OFF HAT, TAKE STETHOSCOPE, WEAR HAT, WEAR STETHOSCOPE, LISTEN TO DIAL, TURN DIAL RIGHT, RIGHT, LEFT, RIGHT, RIGHT	Whirr, Click, Click, Whirr, Kerchunk.	+3
REMOVE STETHOSCOPE, W, UNLOCK BOX 600 WITH KEY, TAKE TOPAZ, E, S	Holmes finds Indian turban muslin. Outside, he is kidnapped.	+5

13. Mycroft (+4)

Command	What happens / why	Pts
W, W, W, S, W, ASK BUTLER ABOUT MYCROFT, GIVE RING TO BUTLER	Mycroft gives the Tower password (randomised; <i>Cleves</i> with seed 42).	+1
E, N, E, E, E, SE, SE, S, E, E, E, SAY CLEVES	Wiggins now follows you.	+3

14. The Tower of London (+8)

Command	What happens / why	Pts
N, N, SE, U, TAKE MACE	The thief's weapon, in the Jewel Room.	+1
D, NW, NE, HIT BUNG WITH MACE, WIGGINS, GET IN BUTT	The Duke of Clarence's butt of malmsey drains, and Wiggins climbs in for the garnet.	+5
OUT, S, S, S, TAKE PADDLE		+1
N, N, N, E, D, LIGHT LAMP, WEAR ARMOUR, U, W, S, S, S, PULL CHAIN	Only Watson in a suit of armour is heavy enough to raise the portcullis.	+1
S, REMOVE ARMOUR, DROP MACE, GET IN BOAT, RAISE ANCHOR, LAUNCH BOAT, PADDLE W x3, PADDLE N, GET OUT OF BOAT	The guards won't let anyone leave by the gate, so leave by water.	

15. The Ampoule (+0)

Command	What happens / why	Pts
N, W, SW, N, W, W, N, W, U, N, W, TAKE AMPOULE, TAKE OFF HAT, PUT AMPOULE IN HAT, WEAR HAT	Holmes's etherium knockout gas. Akbar will take everything you carry, but not your hat.	
E, S, D, E, S, E, E, E, E, E, SE, SE, S, D, W	Swan Lane and the Bar of Gold. Wiggins waits outside.	

16. Akbar (+5)

Command	What happens / why	Pts
WAIT UNTIL 6:00 / 10:00 / 2:00 / 6:00 / 10:00 / 1:50, LOOK, WAIT UNTIL 2:00	The emerald says 20/6/87 and the sapphire 2:00 a.m., which is Monday morning.	
SAY SWORDFISH, GIVE GARNET TO AKBAR	The opal's password, the ruby's carnation, and the garnet's "Give me to Akbar". You are led blindfold to the lair.	+5

17. Moriarty's Lair (+0)

Command	What happens / why	Pts
TAKE OFF HAT, TAKE AMPOULE, HOLD BREATH, BREAK AMPOULE	Holmes holds his breath too. Moriarty and Akbar pass out.	
UNTIE HOLMES, TIE MORIARTY WITH ROPE, TIE AKBAR WITH ROPE	Do this before your breath runs out. You wake at 8:30 with both villains still tied up.	

18. The Coronation (+16)

Command	What happens / why	Pts
TAKE JEWELS, TAKE WHISTLE, TAKE KEY, UNLOCK DOOR WITH KEY, OPEN DOOR, N		+10
BLOW WHISTLE, BLOW WHISTLE, GET IN CAB, BUCKINGHAM PALACE	<i>One blast brings a clueless growler. Two bring a hansom, which arrives at 8:54.</i>	+5
GET OUT OF CAB, GIVE JEWELS TO GUARD	<i>The Queen steps onto the balcony at 9:00 wearing the full regalia.</i>	+1

Where the Points Come From

Event	Pts	Event	Pts
Newspaper	1	Bligh's oar	1
Rousing Holmes	5	Stethoscope	1
Lamp	3	Flower girl revived	5
Magnifying glass	1	Opal	5
Pipe	1	Holmes's ring	1
Tobacco	1	Past the bank guard	3
Matchbook	1	Vault opened	3
Crayon	1	Topaz	5
Pacquet of paper	1	Mycroft's password	1
Invisible ink revealed	5	Into the Tower	3
Haggled telescope	1	Mace	1
Pigeon	1	Garnet	5
Ruby	5	Paddle	1
Safety deposit key	1	Portcullis raised	1
Sapphire	5	Akbar's lair	5
Cigar ash	1	Crown Jewels	10
Torch lit	4	Hansom cab	5
Emerald	5	Regalia delivered	1

Opcode Census

Every instruction executed during the run was decoded with the interpreter's own disassembler and counted by mnemonic. The run used **72 distinct opcodes** over **988,543** instructions. As a version 5 story, Sherlock uses the V5 call family (`call_1n`, `call_2s`, `call_vn2`, ...), `aread` with a timed-input routine that drives its real-time clock, `read_char` for its keypress prompts, and `save_undo`. After the win the game waits at its Quit/Restart/Undo menu, which is where the tally stops.

#	Opcode	Form:No.	Executions	Purpose
1	jz	1OP:128	241,069	branch if zero
2	je	2OP:1	103,732	branch if equal to any operand
3	loadb	2OP:16	57,204	load byte from array
4	store	2OP:13	53,651	store to variable
5	test_attr	2OP:10	52,052	branch if object has attribute
6	add	2OP:20	36,637	signed addition
7	loadw	2OP:15	35,356	load word from array
8	jump	1OP:140	33,621	unconditional jump
9	jg	2OP:3	28,863	branch if greater (signed)
10	call_2s	2OP:25	26,914	call routine with 1 arg, store
11	get_prop_addr	2OP:18	26,308	address of property data
12	get_child	1OP:130	21,732	first child object, branch if any
13	inc	1OP:133	20,991	increment variable
14	get_sibling	1OP:129	19,449	next sibling, branch if any
15	jl	2OP:2	17,125	branch if less (signed)
16	call_vs	VAR:224	14,081	call routine with up to 3 args, store
17	scan_table	VAR:247	12,098	search a table (V4+)
18	div	2OP:23	10,681	signed division
19	test	2OP:7	10,592	branch if all bitmap flags set
20	inc_chk	2OP:5	10,213	increment and branch if greater
21	storew	VAR:225	10,166	store word into array
22	get_prop_len	1OP:132	10,080	length of property data
23	rtrue	0OP:176	9,851	return true
24	rfalse	0OP:177	9,840	return false
25	ret	1OP:139	8,936	return value
26	mul	2OP:22	8,087	signed multiplication
27	print_char	VAR:229	7,653	print ZSCII character

#	Opcode	Form:No.	Executions	Purpose
28	storeb	VAR:226	7,225	store byte into array
29	jin	2OP:6	6,750	branch if object is a child of
30	call_vn	VAR:249	5,975	call routine with up to 3 args, discard result
31	get_parent	1OP:131	5,571	parent object
32	call_vs2	VAR:236	5,504	call routine with up to 7 args, store
33	check_arg_count	VAR:255	5,179	branch if argument N was supplied
34	ret_popped	0OP:184	5,150	return top of stack
35	call_1s	1OP:136	4,732	call routine, no args, store result
36	call_2n	2OP:26	3,667	call routine with 1 arg, discard result
37	call_1n	1OP:143	3,177	call routine, no args, discard result
38	sub	2OP:21	3,159	signed subtraction
39	push	VAR:232	3,024	push onto stack
40	get_prop	2OP:17	2,780	read property value
41	copy_table	VAR:253	2,683	copy or zero a block of memory
42	mod	2OP:24	2,569	signed remainder
43	dec_chk	2OP:4	2,316	decrement and branch if less
44	and	2OP:9	2,163	bitwise AND
45	new_line	0OP:187	1,892	print newline
46	print	0OP:178	1,797	print inline string
47	log_shift	EXT:2	1,781	logical bit shift
48	print_paddr	1OP:141	1,738	print string at packed address
49	output_stream	VAR:243	1,634	select/deselect output streams (memory stream)
50	print_num	VAR:230	1,628	print signed number
51	set_text_style	VAR:241	1,422	bold / reverse / italic / roman
52	set_cursor	VAR:239	1,406	position cursor in upper window
53	set_window	VAR:235	990	select upper or lower window
54	insert_obj	2OP:14	835	move object into another
55	print_table	VAR:254	812	print a rectangle of text
56	put_prop	VAR:227	680	write property value
57	set_attr	2OP:11	635	set object attribute
58	print_obj	1OP:138	571	print object short name
59	clear_attr	2OP:12	522	clear object attribute

#	Opcode	Form:No.	Executions	Purpose
60	save_undo	EXT:9	384	snapshot state for UNDO
61	aread	VAR:228	381	read a line of input (V5, timed, stores terminator)
62	call_vn2	VAR:250	298	call routine with up to 7 args, discard result
63	print_ret	0OP:179	292	print inline string, newline, return true
64	dec	1OP:134	81	decrement variable
65	read_char	VAR:246	61	read a single keypress (V4+)
66	or	2OP:8	33	bitwise OR
67	sound_effect	VAR:245	26	bleeps and sound control
68	remove_obj	1OP:137	22	detach object from parent
69	print_addr	1OP:135	6	print string at byte address
70	random	VAR:231	5	random number / reseed
71	split_window	VAR:234	3	size the upper window
72	erase_window	VAR:237	2	clear a window

Full Command List

One line per input, exactly as replayed by the test (seed 42). Read down each column.

1 z	101 west	201 take emerald	301 say cleves
2 verbose	102 ask sherman for pigeon	202 east	302 north
3 knock on door	103 east	203 south	303 north
4 west	104 north	204 take lamp	304 southeast
5 up	105 west	205 take matchbook	305 up
6 north	106 west	206 east	306 take mace
7 take newspaper	107 south	207 south	307 down
8 give newspaper to holmes	108 look at ruby through telescope	208 south	308 northwest
9 wait	109 show ruby to pigeon	209 south	309 northeast
10 n	110 pigeon, get ruby	210 east	310 hit bung with mace
11 read paper	111 release pigeon	211 light lamp	311 wiggins, get in butt
12 wait	112 north	212 down	312 out
13 n	113 east	213 take oar	313 south
14 west	114 east	214 up	314 south
15 take lamp	115 south	215 turn off lamp	315 south
16 take glass	116 west	216 west	316 take paddle
17 east	117 ask sherman about ruby	217 north	317 north
18 take knife	118 east	218 northeast	318 north
19 take pipe	119 north	219 north	319 north
20 take tobacco	120 east	220 drop crayon	320 east
21 south	121 give shilling to wiggins	221 drop matchbook	321 down
22 down	122 wiggins, steal keys	222 drop telescope	322 light lamp
23 north	123 west	223 take off hat	323 wear armour
24 take matchbook	124 west	224 take stethoscope	324 up
25 south	125 west	225 wear hat	325 west
26 open door	126 south	226 wear stethoscope	326 south
27 east	127 south	227 listen to girl	327 south
28 light lamp	128 south	228 open bag	328 south
29 south	129 southeast	229 open brown bottle	329 pull chain
30 south	130 up	230 take yellow pill	330 south
31 south	131 put white paper in pacquet	231 put yellow pill in girl's mouth	331 remove armour
32 southeast	132 put yellow paper in pacquet	232 remove stethoscope	332 drop mace
33 southeast	133 put blue paper in pacquet	233 south	333 get in boat
34 east	134 open bag	234 east	334 raise anchor
35 east	135 open blue bottle	235 south	335 launch boat
36 k	136 take cotton balls	236 get in boat	336 paddle west
37 read sign	137 put cotton balls in ears	237 pull chain	337 paddle west
38 knock on door	138 wait	238 launch boat	338 paddle west
39 open door	139 wait	239 look	339 paddle north
40 wait	140 wait	240 look	340 get out of boat
41 wait	141 wait	241 drop anchor	341 north
42 wait	142 wait	242 wait until 8:40	342 west
43 n	143 take sapphire	243 y	343 southwest
44 wait	144 wait	244 light lamp	344 north
45 wait	145 n	245 take moss	345 west
46 wait	146 take sapphire	246 put bligh's oar in oarlock	346 west
47 open door	147 wait	247 pull chain	347 north
48 east	148 n	248 look	348 west
49 look	149 wait	249 row north	349 up
50 south	150 n	250 launch boat	350 north
51 southeast	151 wait	251 row west	351 west
52 take crayon	152 n	252 row west	352 take ampoule
53 take pacquet	153 wait	253 row west	353 take off hat
54 turn off lamp	154 n	254 row north	354 put ampoule in hat
55 northwest	155 wait	255 get out of boat	355 wear hat
56 north	156 n	256 east	356 east
57 open pacquet	157 wait	257 east	357 south
58 take white paper	158 n	258 north	358 down
59 put white paper on tomb	159 wait	259 northwest	359 east
60 rub white paper with crayon	160 n	260 northwest	360 south
61 take white paper	161 wait	261 examine ruby with glass	361 east
62 northeast	162 n	262 give ruby to guard	362 east
63 east	163 wait	263 give sapphire to guard	363 east
64 hold white paper over candles	164 n	264 give emerald to guard	364 east
65 turn white paper over	165 wait	265 give opal to guard	365 east
66 south	166 n	266 north	366 southeast
67 east	167 wait	267 take off hat	367 southeast
68 take yellow paper	168 n	268 take stethoscope	368 south
69 put yellow paper on tomb	169 take cotton balls	269 wear hat	369 down
70 rub yellow paper with crayon	170 put cotton balls in blue bottle	270 wear stethoscope	370 west
71 take yellow paper	171 close blue bottle	271 listen to dial	371 wait until 6:00
72 south	172 close bag	272 turn dial right	372 wait until 10:00
73 west	173 down	273 turn dial right	373 wait until 2:00
74 take blue paper	174 northwest	274 turn dial left	374 wait until 6:00
75 put blue paper on henry's tomb	175 north	275 turn dial right	375 wait until 10:00
76 rub blue paper with crayon	176 north	276 turn dial right	376 wait until 1:50
77 take blue paper	177 northeast	277 remove stethoscope	377 look
78 east	178 east	278 west	378 wait until 2:00
79 northwest	179 south	279 unlock box 600 with key	379 say swordfish
80 north	180 look	280 take topaz	380 give garnet to akbar
81 hold yellow paper over candles	181 north	281 east	381 take off hat
82 turn yellow paper over	182 west	282 south	382 take ampoule
83 hold blue paper over candles	183 southwest	283 west	383 hold breath
84 turn blue paper over	184 north	284 west	384 break ampoule
85 west	185 north	285 west	385 untie holmes
86 southwest	186 west	286 south	386 tie moriarty with rope
87 west	187 put tobacco in pipe	287 west	387 tie akbar with rope
88 northeast	188 open matchbook	288 ask butler about mycroft	388 take jewels
89 west	189 take match	289 give ring to butler	389 take whistle
90 haggle with vendor	190 light match	290 east	390 take key
91 haggle with vendor	191 light tobacco with match	291 north	391 unlock door with key
92 buy telescope	192 drop matchbook	292 east	392 open door
93 east	193 drop lamp	293 east	393 north
94 north	194 north	294 east	394 blow whistle
95 north	195 west	295 southeast	395 blow whistle
96 look at statue through telescope	196 take torch	296 southeast	396 get in cab
97 north	197 light clue paper with pipe	297 south	397 buckingham palace
98 east	198 light torch with clue paper	298 east	398 get out of cab
99 east	199 take head	299 east	399 give jewels to guard
100 south	200 melt head with torch	300 east	